

Project Report

Project Name : FitBuddy

1. INTRODUCTION

1.1 Project Overview

FitBuddy is an AI-powered personal health and fitness companion designed to assist users in tracking daily workouts, calculating BMI, managing diet plans, and monitoring progress seamlessly.

1.2 Purpose

The main purpose of FitBuddy is to make personalized fitness guidance accessible and automated for college students and busy working professionals.

2. IDEATION PHASE

2.1 Problem Statement

Many individuals struggle with maintaining a healthy lifestyle due to generic fitness advice, lack of personalized tracking, and expensive personal coaching.

2.2 Empathy Map Canvas

Users feel overwhelmed by unorganized routine tracking, fear falling out of shape, and desire an easy-to-use, smart daily health assistant.

2.3 Brainstorming

Brainstormed solutions prioritized core capabilities: BMI Math Engine, Personalized Diet Generation, Step Tracker Dashboard, and Custom Exercise Routines.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

User Journey: Registration → Enter Height/Weight → View BMI & Daily Target → Access Custom Workout & Diet → Track Daily Progress on Dashboard.

3.2 Solution Requirement

Functional requirements include user authentication, real-time BMI math engine, meal recommendations, and responsive UI dashboards.

3.3 Data Flow Diagram

Data flows from User Interface (Input) → API Processing Server → Machine Learning / Math Engine → Firebase Database → Responsive Dashboard (Output).

3.4 Technology Stack

- **Frontend:** HTML5, CSS3, JavaScript / React
- **Backend:** Python (Flask / FastAPI)
- **Database:** Firebase / SQLite
- **Analytics:** Scikit-Learn

4. PROJECT DESIGN

4.1 Problem Solution Fit

FitBuddy directly solves generic guidance issues by delivering automated, user-specific calorie and exercise targets tailored to individual goals.

4.2 Proposed Solution

An intuitive mobile-first web app that integrates personal health profiles with algorithmic fitness suggestions.

4.3 Solution Architecture

Modular architecture separating Presentation Layer (Web UI), Logic Layer (Python REST API), and Storage Layer (Cloud Database).

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Executed across 4 Sprints: Sprint 1 (Auth & Profile), Sprint 2 (BMI & Calorie Logic), Sprint 3 (Workout & Diet Engine), Sprint 4 (Dashboard & Testing).

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Validated input fields, API connection, meal generation logic, and verified response time remains under 2 seconds across active scenarios.

7. RESULTS

7.1 Output Screenshots

[FitBuddy Health Dashboard, BMI Calculator Interface, and Calorie Progress Chart UI components integrated successfully].

8. ADVANTAGES & DISADVANTAGES

Advantages: Personalized guidance, zero subscription cost, user-friendly UI, instant updates.

Disadvantages: Requires active internet connection for real-time recommendations.

9. CONCLUSION

FitBuddy effectively bridges the gap between complex health data and daily wellness, delivering an accessible lifestyle platform for users.

10. FUTURE SCOPE

Future enhancements include real-time pose estimation via computer vision for posture correction and wearable smartwatch integration.

11. APPENDIX

Source Code: GitHub Repository Available
Dataset Link: Open Health & Fitness Dataset
GitHub & Project Demo Link: <https://github.com/fitbuddy-demo/project>